Introduction to Quantum Chemistry

- Classical Mechanics — Everyday sized objects.
- Quantum Mechanics — Very very small objects.

"Essentially all of chemistry is a consequence of the laws of quantum mechanics"

- $\boxed{\text{Quantum Computer}}$
- Combustions of octane
- $N_2 + O_2 \longrightarrow$
- Eq. constant of a chemical reation.
- Spectroscopy — Abs, Em, Fluorescence, NMR, IR
- Structure of protein molecule.
- Structure and function DNA
  ↳ "Quantum Biology" "Quantum Materials"

Richard Feynman — "Everything that living things do can be understood in terms of jiggling and wiggling of atoms"

Classical	Quantum
large	very small
heavy	light
continuous	discret / quantized
Newton's law	Schrodinger Eq.
Trajectory	wave function
deterministic	probabilistic
intutive	non-intuitive

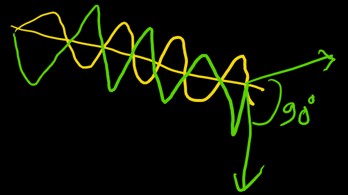
Ref. Books:

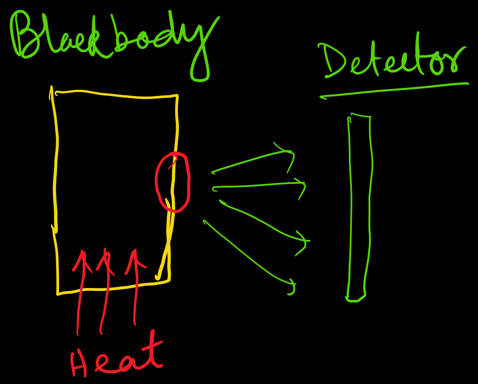
1. Physical Chemistry
   IRa N Levine
2. ✓ Physical Chemistry
   McQuarrie & Simon
3. ✓ Quantum Chemistry — "

Blackbody Radiation

Maxwell's theory of electricity, magnetism, El-mag Radiation, thermodynamics. (before 1900).

- Light is a electromagnetic wave – Classical theory.





$C = \lambda \nu$ ← wavelength, frequency

$C = 2.998 \times 10^{8}\ m/s$

Classical Theory $\rightarrow \rho_\nu(T) = \frac{8\pi k_B T}{c^3}\nu^2$

Rayleigh – Jeans Law

$\longrightarrow$ "Ultraviolet catastrophe"

Quantum Theory

1900 → Max Plank

$E = n h \nu$

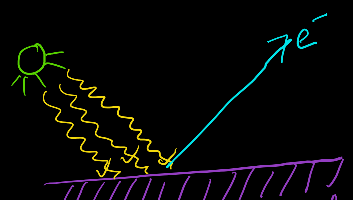
Energy of oscilator (E), Integer (n), Planck Constant (h), Frequency (ν)

$$\rho_\nu(T) = \frac{8\pi h}{c^3}\frac{\nu^3}{e^{h\nu/k_BT}-1}$$

h = 6.626 x 10^-34 J.s

Assumed that energy is quantized

Photoelectric effect



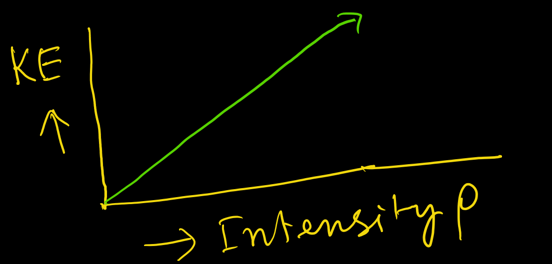
Metal Surface

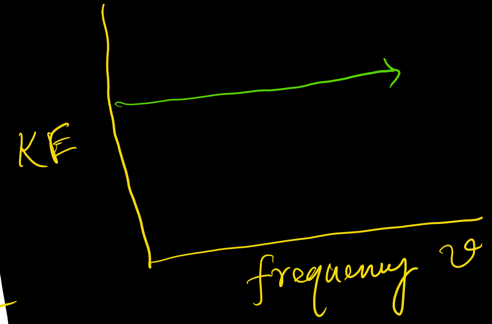
Ejected Electron | $KE = \frac{1}{2}mv^2$

1886 Hertz : UV light removes e^- from metal

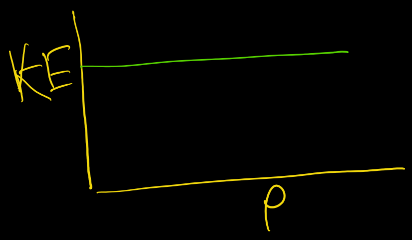
- KE of e^- is independent of Intensity
- Photoelectric effect doesnot happen for any frequency of light
- Requires a threshold frequency ν_0

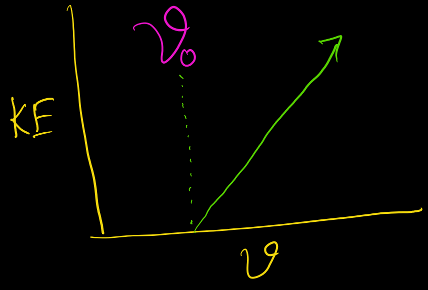
Classical Theory





Quantum Theory





Quantum Theory → Einstein 1905

Photons → $E = nh\nu$ → same 'h'

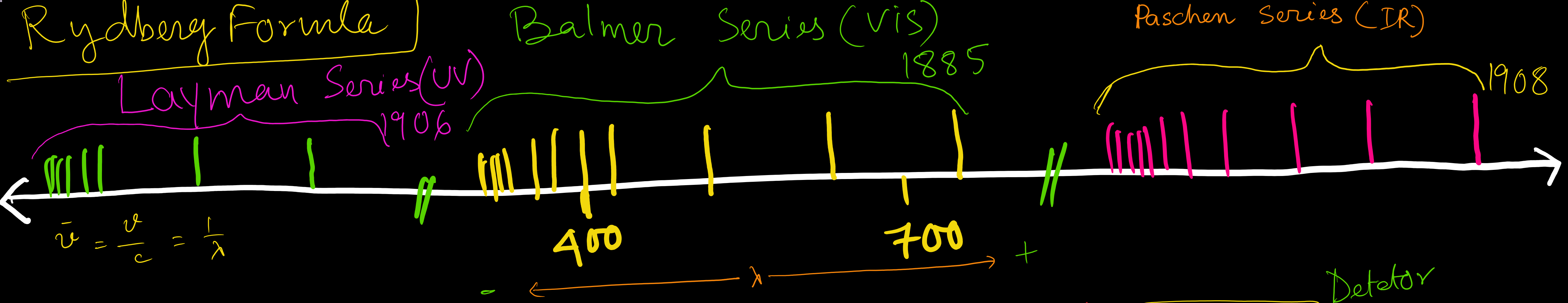
$\nu = \frac{c}{\lambda}$

$KE = h\nu - h\nu_0 = h\nu - \phi$

$\phi = h\nu_0 \Rightarrow$ "Work Function"

Radiation consists of small packet of energy ⟹ "Photons"

Rydberg Formula



$\bar{\nu} = \frac{\nu}{c} = \frac{1}{\lambda}$

$\boxed{R_H = 109,737 \cdot 3 \text{ cm}^{-1}}$ "Rydberg Constant"

$\bar{\nu} = R_H\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$ Lyman $n > 1$ $n \in \mathbb{Z}$

$\bar{\nu} = R_H\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$ Balmer $n > 2$ $n \in \mathbb{Z}$

$\bar{\nu} = R_H\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$ Paschen $n > 3$ $n \in \mathbb{Z}$

$R_{expt} = 109,677 \cdot 6 \text{ cm}^{-1}$

Detetor

$\boxed{\text{Atoms}} \xrightarrow{\text{heat}} \boxed{\text{Radiation}} \rightarrow \boxed{\text{Spectrum}}$

$$\boxed{\bar{\nu} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$

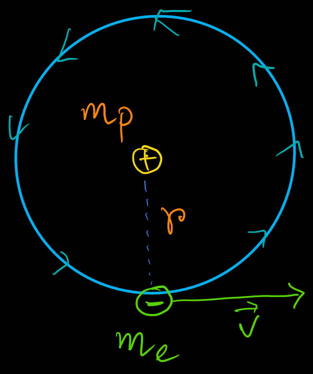
$n_2 > n_1 > 0$

$n_1, n_2 \in \mathbb{Z}$

"Rydberg Formula"

The Bohr Theory of Hydrogen Atom – Radius

Angular vs	Linear Motion
$m \leftrightarrow I$	$I = mr^2$
$v \leftrightarrow \omega$	$v = r\omega = 2\pi r\nu \leftarrow$ freq.
$p \leftrightarrow l$	$p = mv \quad l = I\omega$
$T \leftrightarrow T$	$T = \frac{1}{2}mv^2 = \frac{1}{2}\frac{p^2}{m}$ $T = \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{l^2}{I}$



I = Moment of Inertia
ω = Angular velocity
l = Angular momentum

$\oplus \leftarrow \qquad \ominus \rightarrow$

(Coulomb Force) = (Centrifugal Force)

$$\frac{e^2}{4\pi\varepsilon_0 r^2} = \frac{m_e v^2}{r}$$

ε_0: permitivity of free space

Two unkown

$\downarrow$

v, r

Bohr 1911 → Assumed that angular momentum is quantized.

$$l = m_e v r = n\hbar \qquad n \geq 1,\ n \in \mathbb{Z}$$

'h-bar' $\hbar = h/2\pi$ → Reduced Planck's constant

$$v = \frac{n\hbar}{m_e r}$$

$$\frac{e^2}{4\pi\varepsilon_0 r^2} = \frac{m_e}{r}\left(\frac{n\hbar}{m_e r}\right)^2 = \frac{m_e n^2 \hbar^2}{r^3 m_e^2}$$

$$r m_e e^2 = 4\pi\varepsilon_0 n^2 \hbar^2$$

$$r = \boxed{\frac{4\pi\varepsilon_0 \hbar^2 n^2}{m_e e^2}} = 5.29 \times 10^{-11}\,m \ (\text{if } n=1) = 52.9\ pm$$

$$r = 0.529\ \text{Å}, \quad \boxed{1\ \text{Å} = 100\ pm = 10^{-10}\ m}$$

Bohr radius → a_0

The Bohr Hydrogen Model – Energy

$$E = \underset{K.E.}{T} + \underset{P.E.}{V} = \frac{1}{2}m_e v^2 - \frac{e^2}{4\pi\varepsilon_0 r} \quad \left[\begin{smallmatrix} q_1 & q_2 \\ + & - \\ p & e \end{smallmatrix}\right]$$

From prev. slide ⇒

$$\frac{e^2}{8\cancel{4}\pi\varepsilon_0 r^{\cancel{2}}} = \frac{m_e v^2}{2\cancel{r}}$$

$$\boxed{\frac{1}{2}m_e v^2 = \frac{e^2}{8\pi\varepsilon_0 r}}$$

$$E = \frac{-e^2}{8\pi\varepsilon_0 r} = \frac{-e^2}{8\pi\varepsilon_0}\left(\frac{m_e e^2}{4\pi\varepsilon_0 \hbar^2 n^2}\right)$$

$$\boxed{E = \frac{-m_e e^4}{32\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n^2}\right)}$$

n = quantum no.

$$\Delta E = E_{n_2} - E_{n_1} = h\nu = hc\bar{\nu}$$

$$= \frac{m_e e^4}{32\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$$

Similar to Rydberg Formula

$$\nu = \frac{c}{\lambda} \qquad \bar{\nu} = \frac{1}{\lambda}$$

$$\boxed{\bar{\nu} = R_H\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)} \qquad n_1 > n_2 \qquad n_1, n_2 \in \mathbb{Z}$$

$$\boxed{R_H = \frac{m_e e^4}{8\varepsilon_0^2 c h^3}} \qquad R_H = 109,737.3\ cm^{-1}$$

accurate w/in 0.05% of expt.

$m_p \neq \infty \qquad m_e \rightarrow \mu = \left(\frac{1}{m_e} + \frac{1}{m_p}\right)^{-1} = \frac{m_e m_p}{m_e + m_p}$

$m_p = 1.672 \times 10^{-27}$ kg, $m_e = 9.109 \times 10^{-31}$ kg | $\mu = 0.9995\, m_e$

$m_e \rightarrow \mu$, $R_H \rightarrow R_1$ | $R_1 = 109,676\ cm^{-1}$

w/in 0.001% error ($109,677.6\ cm^{-1}$)

Amazing accuracy when Bohr assumed that angular momentum is quantized.

Wave Particle Duality: The De Broglie Hypothesis

Einstein → 1905 → Light is a particle and a wave

De Broglie → 1924 → Matter is too

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

λ → wavelength

p → momentum

If m → Large, $\lambda \approx 0$

If m → Small, $\lambda > 0$

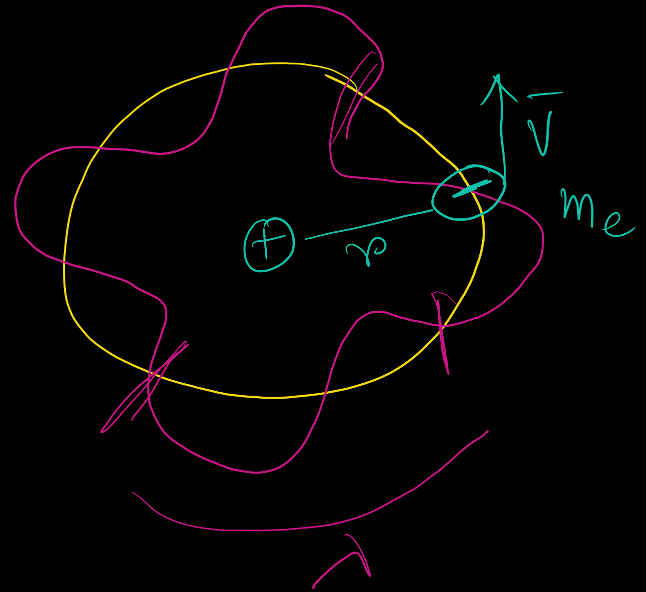
de Broglie waves for electron interference:

$$2\pi r = n\lambda$$

(circumference)

$$= \frac{nh}{p}$$

$$= \frac{nh}{mv}$$



$$mvr = \frac{nh}{2\pi} \Rightarrow \boxed{l = n\hbar}$$

Bohr Quantization Condition.

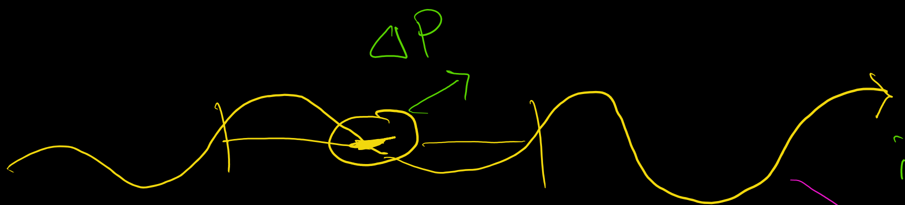
$$\boxed{l = mvr}, \quad \hbar = \frac{h}{2\pi}$$

angular Momentum

Heisenberg Uncertainity Principle

Position and momentum of a particle can not be specified simultaniously with unlimited precision

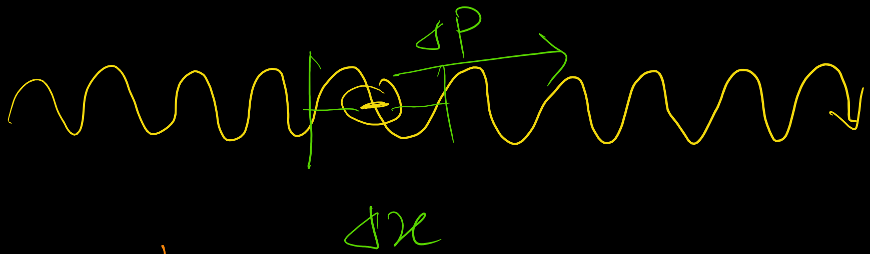
$\Delta x, \Delta P$ → uncertainity in x, P



Large λ of photon

$\Delta x \approx \lambda_{photon}$

Small λ of photon



$P = \frac{h}{\lambda} \longrightarrow \lambda\downarrow = P\uparrow \qquad \Delta x\downarrow = \Delta P\uparrow$

$$\boxed{\Delta x \, \Delta P \geq \hbar/2}$$

1920's

Heisenberg's Uncert principle

Macroscopic → Not Important

Microscopic → Important

Bhor Model ⇒ $\Delta x = \Delta P = 0$ ⇒ Bad!

Limitation of Bohr's H-Model

Need more general theory.